ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594
2. Isolating specific categories first can break down the data, reduce the sample size during training, thereby improving model accuracy and enabling targeted analysis.

AI used for this answer: Yes / No

If yes, how: No

External references (links):

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

Other than the simple case where the category column just wasn't mapped or added to a new category_name column, using dropna() here is flawed because losing 40% of the data causes severe information loss. Additionally, dropna() fails to catch pseudo-nulls like string 'NaN', 'None', or empty strings (') without manual cleaning.

An alternative method is to first use .replace() to convert these misformatted strings into true nulls, and then treat the missing records as an "Unknown" category (or use clustering) so that the valuable text data can still be utilized.

AI used for this answer: Yes / No

If yes, how: Yes. Translate Chinese expressions into formal academic phrasing.

External references (links): https://vocus.cc/article/685171b2fd89780001ccebe3

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. No. The trends are similar but not exactly the same.
2. It is highly likely that categories with a very small number of records will not be sampled proportionally during random selection, leading to severe statistical bias.

AI used for this answer: Yes / No

If yes, how: Yes. For translation.

External references (links):

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?

2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. Increases.
2. The overall sparsity will decrease because non-zero counts increase significant by stopword.

AI used for this answer: Yes / No

If yes, how: No

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	**Number of terms kept (keep == True)**
0.0	35482
0.01	5089
0.05	1491

2. **Words that are used either extremely frequently or extremely rarely across various articles.**

3. **Extremely frequent words fail to provide useful distinct features, making it difficult for the model to analyze or differentiate categories in subsequent training. On the other hand, extremely rare words are overly specific to a tiny fraction of the data, which means they cannot effectively help the model generalize or accurately determine the target categories for future articles.**

AI used for this answer: Yes / No

If yes, how: Yes. For translation.

External references (links):

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. Domain-specific terminology for graphics.
2. Pearson considers the linear correlation between the terms' raw counts and a binary class indicator. Meanwhile, Spearman considers the monotonic relationship based on the ranks of the word counts rather than their raw values.
3. When a word's count in that category is drastically higher than in any other category.

AI used for this answer: Yes / No

If yes, how: Yes. To deepen the understanding of the analytical principles behind Pearson and Spearman.

External references (links):

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

The runtime increased from 0.099 seconds to 1.466 seconds. The pattern count increased from 710 to 9,547.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

Top-K k restricts the exact number of top rows returned.

MaxFPGrowth minSup sets the minimum count a pattern must have to be kept.

Answer (3) — Maximal vs standard frequent patterns:

Maximal Frequent Patterns increase information density to successfully avoid data repetition and redundancy.

AI used for this answer: Yes / No

If yes, how: Yes. Organizing thoughts and translation.

External references (links):

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. Visually, human perception is limited to 2D or 3D spaces, making it impossible to directly view high-dimensional data. Mathematically, it triggers the "curse of dimensionality", where data points become nearly equidistant, causing distance-based analysis to lose discriminative power.
2. A category known as Dimensionality Reduction techniques must be implemented, including PCA, t-SNE, and UMAP, to project high-dimensional spaces onto lower-dimensional visual planes while preserving data relationships.

AI used for this answer: Yes / No

If yes, how: Yes. Organizing thoughts and translation.

External references (links):

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. UMAP shows the most distinct, separated cluster islands**.** PCA shows heavy overlap around the center.
2. PCA struggles because it is a strict linear dimensionality reduction technique that preserves global variance. In contrast, t-SNE and UMAP use non-linear, graph-based methods to map these curved neighbors to two-dimensional/three-dimensional space.
3. t-SNE is a local-structure-preserving algorithm that optimizes only to keep true neighbors close together, while discarding global geometric properties. Therefore, the long-range physical distance between separate clusters carries no meaningful statistical significance.

AI used for this answer: Yes / No

If yes, how: Yes. Organizing thoughts and translation.

External references (links):
https://blog.csdn.net/dongke1991/article/details/160169589

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. 4
2. Using numbers as labels misleads the algorithm into assuming a mathematical logic, which consequently results in artificial ordering or incorrect arithmetic operations between categories.
3. One-hot binarization represents categories as vectors like [1, 0, 0, 0] and [0, 1, 0, 0] to map each category into an independent orthogonal dimension, ensuring that all categories are placed at an equal mathematical distance from one another.

AI used for this answer: Yes / No

If yes, how: Yes. Translation

External references (links):
